A Practical Benchmark of CNN, Transformer, and SAM2-Based Segmenters on Pascal VOC 2007

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Abstract

This report studies semantic segmentation on Pascal VOC 2007 under a shared optimization and evaluation pipeline that compares a classical encoder-decoder baseline, a stronger CNN decoder, a transformer segmenter, and a lightweight SAM2-based semantic adapter. Because these systems start from different pretrained sources (ImageNet, ADE20K, and SAM2 pretraining), the study should be read as a pragmatic system-level benchmark rather than as a clean architecture-isolation experiment. The official validation split of 213 images is treated as a locked test set, while the official training split is partitioned into 178 internal training images and 31 internal development images for model selection. Across four representative systems, SegFormer-B2 achieves the best official-validation overlap metrics at 56.0% mIoU, DeepLabV3+ attains the best HD95, and the adapted SAM2-LoRA variant substantially outperforms the frozen SAM2 proof-of-concept. The ablations show that model capacity, augmentation strength, and adapter tuning strategy matter more than swapping among simple loss formulations, and that the old 60-epoch budget would have materially under-trained the strongest U-Net backbone. The resulting system goes beyond the assignment minimum by evaluating eleven trained models, analyzing subset-specific failure modes, and presenting both quantitative and qualitative evidence under a publication-style protocol.

1 Introduction

Semantic segmentation remains a useful stress test for dense prediction because good performance requires both recognition and localization. Pascal VOC 2007 is small enough to make architectural and optimization choices visible, yet varied enough to expose the weaknesses of under-powered decoders, insufficient augmentation, and brittle boundary handling [Everingham et al., 2010]. That combination makes it a good benchmark for a course project that aims to compare families of models rather than merely reproduce one paper.

The project requirements ask for U-Net and SAM, at least one additional model family, quantitative metrics, qualitative examples, and at least two ablations. I extend that baseline in three ways. First, I evaluate four representative headline systems spanning CNN, transformer, and foundation-model design patterns. Second, I run a larger ablation suite over backbone capacity, loss, augmentation, and SAM2 tuning. Third, I use a stricter reporting pipeline with a locked official validation split, figure-specific analysis, and subset-level generalization diagnostics. The goal is not to claim a new state of the art, but to produce a careful small-scale study that explains what actually works under a realistic compute budget. Because the pretrained starting points differ across families, the paper focuses on a practical comparison of deployable systems rather than on a perfectly matched architecture-only ablation.

Preprint.

2 Dataset and Experimental Protocol

Pascal VOC 2007 provides pixel-wise semantic labels for 20 foreground classes plus background [Everingham et al., 2010]. I preserve the official validation split as a final test set and never tune on it. The official training split contains 209 labeled images, which I partition into 178 internal training images and 31 internal development images using a deterministic label-aware split heuristic. This internal development partition is used for early stopping, learning-rate control, and checkpoint selection; all headline tables in the report are reported on the untouched official validation split of 213 images. Because the internal development set is very small and does not represent every class equally, I treat it as a noisy model-selection signal rather than as a second benchmark. In particular, the realized split omits the bicycle class entirely and contains only one or two examples for several other rare classes.

All models operate on ImageNet-normalized RGB inputs and use a 512 pixel resolution during evaluation. Training-time preprocessing depends on the augmentation preset but always maintains paired image-mask geometry, respects the VOC ignore label, and ends with tensor conversion plus normalization. The default *standard* augmentation preset uses random scale jitter, random crop-or-pad to 512, horizontal flips, moderate color jitter, and light Gaussian blur. The ablation suite swaps this preset with *none* and *strong* variants to test whether a small benchmark still benefits from more aggressive appearance perturbations.

In addition to the raw split, I annotate the validation images with four subset tags intended to support targeted failure analysis: person-present (79 images), small-object (117 images), crowded-scene (29 images), and high-boundary-complexity (56 images). These tags are computed from mask geometry rather than hand-written labels, which keeps the subset analysis reproducible and inexpensive, but they should be interpreted as descriptive proxies rather than as canonical semantic regimes.

3 Models

The first headline model is a U-Net-style encoder-decoder with a ResNet encoder, bilinear upsampling, and skip-connected decoder blocks [Ronneberger et al., 2015, He et al., 2016]. This model is intentionally plain: it provides a strong and interpretable baseline, and it is also the base architecture for the backbone, loss, and augmentation ablations. The headline configuration uses a ResNet-34 encoder pretrained on ImageNet.

The second model is DeepLabV3+ with a ResNet-50 backbone and atrous spatial pyramid pooling [Chen et al., 2018]. DeepLabV3+ is included because it remains a strong CNN segmentation baseline and provides a useful reference point for whether a transformer is buying real performance or merely additional complexity.

The third model is SegFormer-B2, loaded from the pretrained ADE20K checkpoint exposed by the Hugging Face implementation [Xie et al., 2021]. SegFormer is attractive here because it combines strong multiscale reasoning with a relatively light decoder, making it a good candidate for a small but diverse dataset like VOC.

The final headline family is a semantic segmentation adapter built on top of the SAM2 Hiera-S image encoder [Ravi et al., 2024]. Rather than using prompt-based mask decoding, the project repurposes SAM2 features for dense semantic prediction via a shallow convolutional decoder. Two variants are studied: a frozen-encoder baseline and a LoRA-enabled version that applies low-rank adapters to attention and MLP submodules inside the image encoder [Hu et al., 2022]. This isolates whether foundation-model features are already strong enough for dense semantic transfer, or whether lightweight task adaptation is essential.

The pretrained starting points differ materially across families: U-Net and DeepLabV3+ use ImageNet ResNet weights, SegFormer uses an ADE20K semantic-segmentation checkpoint, and SAM2 starts from the published SAM2 image encoder. This mismatch is acceptable for a practical benchmark, but it means the headline comparison conflates architecture choice with the strength of the transfer prior.

4 Training Setup

All experiments share the same optimization policy. Training uses AdamW with model-specific learning rates, a plateau scheduler with a maximum budget of 150 epochs, learning-rate reduction patience of five validation epochs, and early stopping patience of 15 validation epochs after a minimum of 20 epochs [Loshchilov and Hutter, 2019]. Validation mIoU is the sole monitored quantity for checkpointing, learning-rate scheduling, and stopping decisions. This policy deliberately treats the epoch limit as a safety ceiling rather than as the definition of convergence. Consistent with that setup, I use official-validation mIoU as the primary ranking metric in the headline comparison, while Dice, pixel accuracy, and HD95 are treated as secondary diagnostics.

The default loss is a weighted sum of cross-entropy and Dice, with two U-Net ablations that replace it by pure cross-entropy or pure Dice. Batch sizes depend on memory footprint: U-Net runs at batch size 8 except for the ResNet-50 variant, DeepLabV3+ uses batch size 4, and the transformer- and SAM2-based models use batch size 1 with gradient accumulation where needed. Mixed precision is disabled on MPS because the relevant kernels were less reliable than full precision in this environment. All evaluation metrics are computed on a common 512×512 resize-and-pad canvas with padded pixels ignored. HD95 is averaged over foreground class-image pairs in which the class is present, which makes it a useful boundary-sensitive diagnostic but a noisier aggregate than mIoU.

All training was performed on a MacBook Pro with an Apple M5 Max and 64 GB unified memory. That hardware is not a conventional segmentation workstation, but it is representative of the kind of single-device environment in which a course project is actually executed. Reporting training hours and evaluation throughput is therefore not incidental; it is part of the practical comparison.

5 Main Results

Table 1 gives the official-validation comparison for four representative systems: U-Net-ResNet34, DeepLabV3+, SegFormer-B2, and SAM2-LoRA. I use the adapted SAM2 variant here because it is the strongest member of that family; the frozen version is retained below as an ablation and transfer probe. SegFormer-B2 is the strongest model by the primary overlap metrics, with the best mIoU, Dice, and pixel accuracy. DeepLabV3+ is the runner-up on mIoU and also achieves the best HD95, which means the ranking depends somewhat on whether one prioritizes region overlap or boundary distance. SAM2-LoRA closes much of the gap to DeepLabV3+ and clearly surpasses the plain U-Net baseline, but it still trails the best purpose-built segmenters.

Table 1: Official-validation results for the four representative systems. All metrics are computed on the locked VOC 2007 validation split. mIoU is the primary ranking metric; HD95 is a secondary boundary diagnostic computed on the common 512×512 evaluation canvas and averaged over foreground class-image pairs.

Model	mIoU	Dice	Pixel Acc.	HD95 ↓	Train hrs	Infer FPS	Params (M)
SegFormer-B2	**56.0**	**69.6**	**90.0**	186.8	0.9	**3.6**	27.4
DeepLabV3+	46.4	60.4	85.6	**148.4**	0.6	3.1	42.4
SAM2-LoRA	44.2	57.5	87.7	245.0	0.8	3.6	48.0
U-Net-R34	23.7	33.7	81.9	269.1	1.9	3.1	29.9

Figure 1 shows the corresponding accuracy-runtime trade-off across all eleven experiments, not just the four headline models. Two patterns are worth noting. First, SegFormer-B2 sits on the best practical frontier in this study: it is not the absolute fastest model to train, but no other experiment combines its accuracy with a similar runtime. Second, the strongest U-Net variant requires substantially more training time than its smaller counterparts before it pays off, which foreshadows the backbone ablation result that the old 60-epoch cap would have been too conservative.

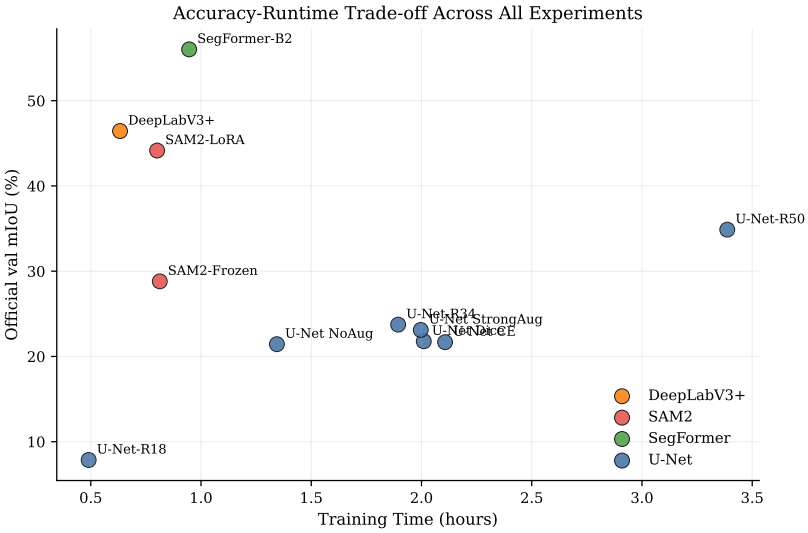

Figure 1: Official-validation mIoU versus end-to-end training time for all completed experiments. Each point corresponds to one trained model under the common training policy.

Figure 2 breaks the headline comparison down by class. The overall ranking is not driven by one easy category. SegFormer-B2 is broadly strong across animal and vehicle classes, which suggests that its gain comes from more stable multiscale reasoning rather than from overfitting to one foreground type. DeepLabV3+ remains competitive on several rigid object classes and often looks cleaner under the boundary metric. SAM2-LoRA improves substantially over the frozen adapter and is often much closer to DeepLabV3+ than to U-Net, but it is still inconsistent on thin or infrequent categories. U-Net-ResNet34 is the weakest baseline across most classes.

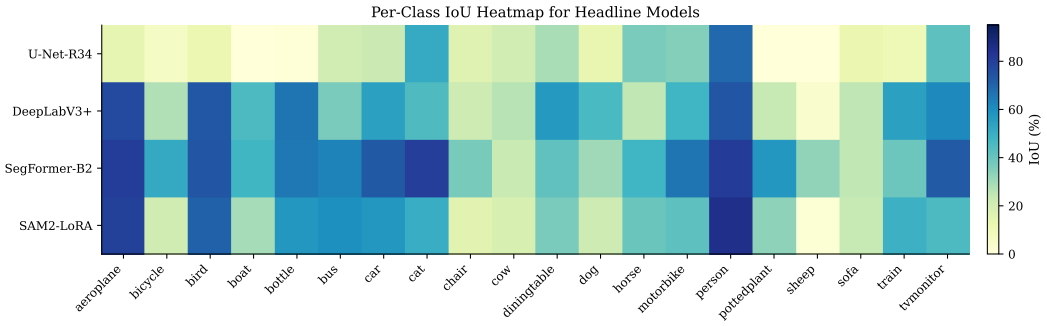

Figure 2: Per-class IoU heatmap for the headline models, excluding background. SegFormer-B2 is consistently strong across most classes rather than winning through a single outlier category.

6 Ablation Studies

Table 2 summarizes the ablation suite. The backbone study is the clearest result: scaling the U-Net encoder from ResNet-34 to ResNet-50 provides a large accuracy gain, while shrinking to ResNet-18 collapses performance. That asymmetry suggests the baseline U-Net was primarily capacity-limited rather than over-regularized. The loss ablation is subtler. Pure Dice and pure cross-entropy both underperform the default mixed objective, which is consistent with the idea that cross-entropy stabilizes class competition while Dice improves overlap on sparse foreground regions. Augmentation also matters, but not in a monotone way: removing augmentation hurts materially, while the stronger preset recovers most of that loss without surpassing the default standard policy.

Table 2: Official-validation ablation results. The delta column is measured relative to the default baseline within each group.

Group	Variant	mIoU	Δ mIoU	Dice	Train hrs
Backbone					
Backbone	U-Net-R18	7.9	-15.9	9.6	0.5
Backbone	U-Net-R34	23.7	+0.0	33.7	1.9
Backbone	U-Net-R50	34.9	+11.1	48.6	3.4
Loss					
Loss	U-Net CE	21.7	-2.1	30.5	2.1
Loss	U-Net-R34	23.7	+0.0	33.7	1.9
Loss	U-Net Dice	21.8	-2.0	31.6	2.0
Augmentation					
Augmentation	U-Net NoAug	21.4	-2.3	30.6	1.3
Augmentation	U-Net-R34	23.7	+0.0	33.7	1.9
Augmentation	U-Net StrongAug	23.1	-0.6	32.7	2.0
SAM2 adaptation					
SAM2 adaptation	SAM2-Frozen	28.8	+0.0	41.0	0.8
SAM2 adaptation	SAM2-LoRA	44.2	+15.3	57.5	0.8

The SAM2 tuning ablation is especially informative. The frozen adapter is a useful proof of concept, but LoRA improves it by 15.3 mIoU points, which is one of the largest gains in the entire study. In other words, SAM2 features are not enough on their own; lightweight task adaptation is the difference between an interesting transfer baseline and a genuinely competitive model. For that reason, SAM2-LoRA rather than SAM2-Frozen is used as the representative SAM2 system in the main comparison.

Figure 3 explains why a shared 60-epoch budget would have been misleading. U-Net-ResNet18 saturates early, and U-Net-ResNet34 plateaus on a moderate timescale, but U-Net-ResNet50 keeps climbing well after epoch 60. This is precisely the behavior that motivated the training-policy change: a hard cap that looks harmless for smaller models can still truncate the most promising one.

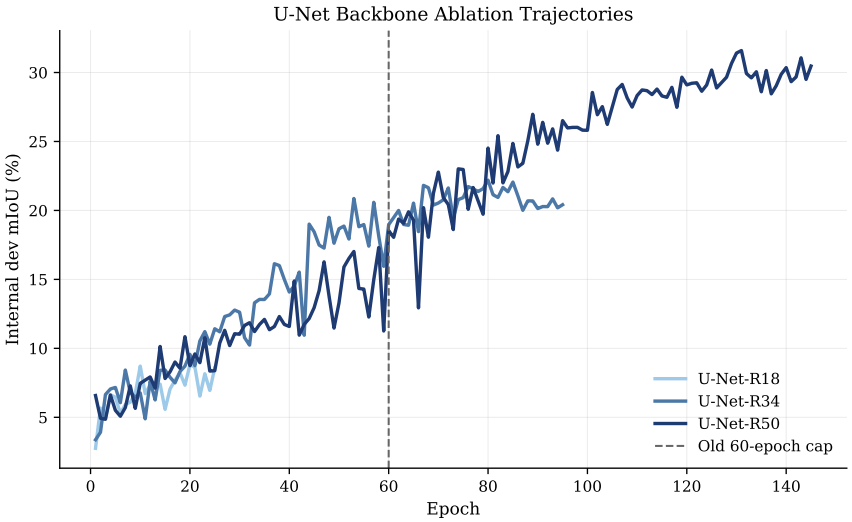

Figure 3: Internal-development mIoU trajectories for the U-Net backbone ablation. The dashed vertical line marks the old 60-epoch ceiling. ResNet-50 continues to improve well beyond that point.

7 Generalization and Failure Analysis

Figure 4 reports mIoU over the four tagged validation subsets in addition to the aggregate score. Because these subset tags are geometry-derived proxies rather than curated semantic regimes, I use them for descriptive failure analysis rather than for strong causal claims. The crowded-scene subset is nevertheless the hardest regime for every model in the study, which is unsurprising because crowded masks combine class ambiguity, occlusion, and more boundary transitions. Small-object and high-boundary-complexity cases are also substantially harder than the overall average. The encouraging result is that the overall mIoU ranking remains fairly stable across subsets: the best global models are also the most reliable specialized models. The disappointing result is that no model closes the gap on crowded scenes, so this remains the clearest open problem in the project.

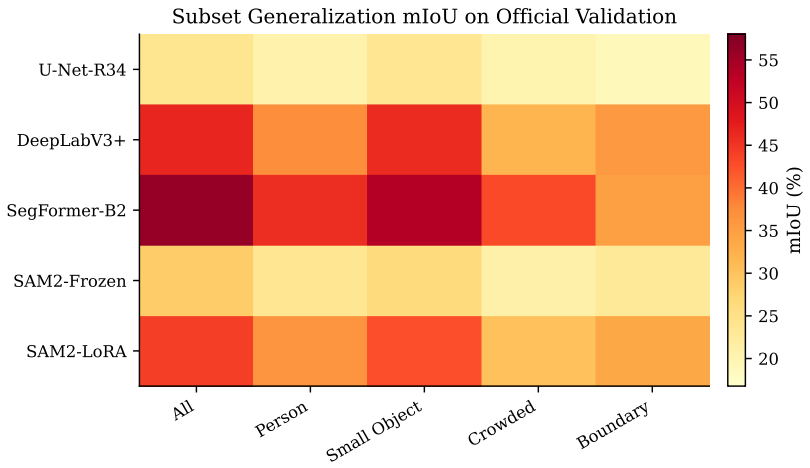

Figure 4: Subset-specific official-validation mIoU. Crowded scenes remain the hardest setting for every model, while SAM2 benefits dramatically from LoRA adaptation on the same subsets.

Figure 5 makes the ranking more concrete. The easy row shows that all models can solve a clean single-object scene once the foreground occupies enough pixels. The crowded and articulated-person rows are more revealing. U-Net tends to fragment masks or miss object extent entirely, DeepLabV3+ usually recovers the right coarse layout but can blur fine boundaries, and SegFormer-B2 tends to maintain the best compromise between global context and local structure. SAM2-LoRA is substantially stronger than the frozen adapter, but it still tends to miss fine extremities or exact object extent under clutter.

The SAM2-specific comparison in Figure 6 shows what the LoRA gain actually looks like. The frozen model often identifies the correct object family but undershoots extent and topology. LoRA closes that gap by refining boundaries and recovering missed regions without changing the lightweight decoder. The qualitative effect is consistent with the quantitative jump in Table 2: the adapter is not just improving average overlap numerically; it is correcting visible segmentation failures.

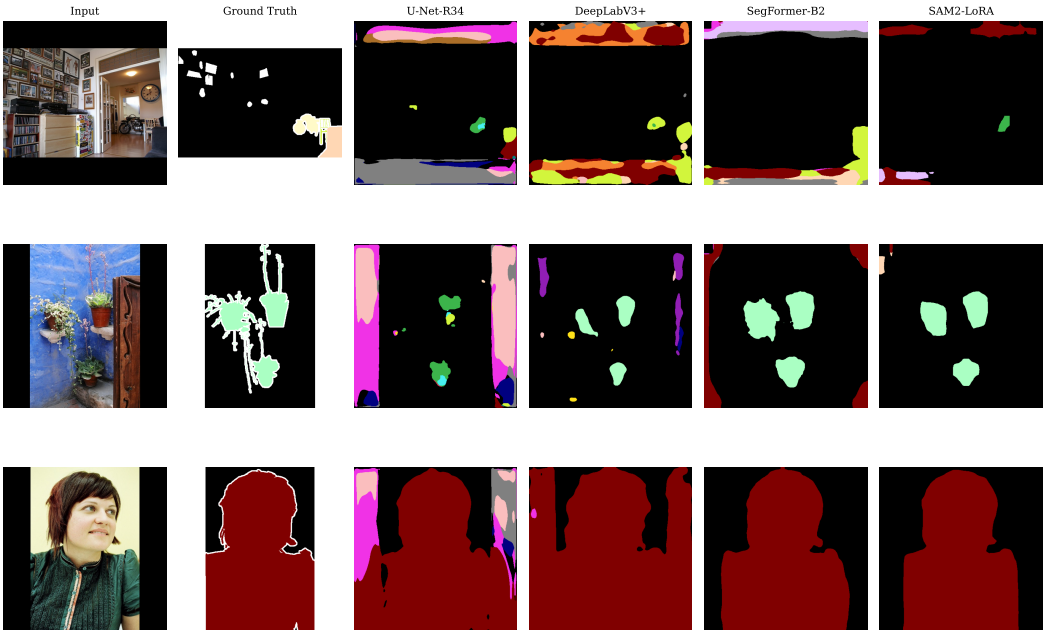

Figure 5: Qualitative comparison on representative official-validation images selected automatically from the headline runs. The rows cover a crowded scene, a boundary- or person-sensitive case, and an easy high-confidence example.

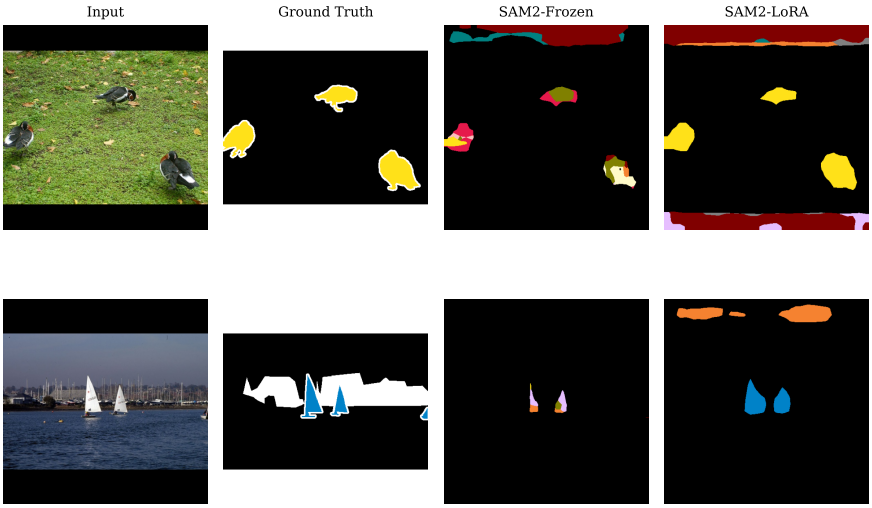

Figure 6: Representative SAM2 comparisons on images where LoRA produces the largest per-image gains. The improvement is primarily in extent recovery and boundary refinement rather than in coarse object discovery.

8 Lessons Learned and Limitations

Several lessons stand out. First, pretrained transformer features are extremely effective even on a relatively small benchmark, provided the decoder is designed for dense prediction. Second, classical CNN systems remain competitive: DeepLabV3+ is not far behind SegFormer-B2 on mIoU and is actually strongest on HD95, so the preferred model depends on whether region overlap or boundary distance is the primary objective. Third, the right training policy matters almost as much as the architecture itself. The backbone ablation would have been misread under the old fixed 60-epoch cap,

Figure 7: Top-three best and worst SegFormer-B2 triptychs on the official validation split. The strong examples are mostly single-object scenes, while the weak ones involve clutter, thin boundaries, or ambiguous extent.

and the LoRA result would have been easy to miss if SAM2 were evaluated only as a frozen feature extractor.

The project also has clear limitations. VOC 2007 is small, so architecture ranking may be more sensitive to the inductive biases of the benchmark than to true large-scale robustness. More importantly, the pretrained initializations differ across families, so the headline comparison should be read as a pragmatic benchmark of end-to-end systems rather than as a clean architecture-only study. The official validation split is treated correctly as a test set, but there is still only one split, and the internal development split is only 31 images, missing bicycle entirely and representing several classes only once. The hardware environment also shapes what is practical: larger transformer variants or full SAM2 fine-tuning were deliberately excluded because they would not have been a responsible use of time on the available device. Finally, although the report includes subset and qualitative analysis, it does not yet include repeated seeds, confidence intervals, or an original-resolution boundary study; all three would strengthen the statistical side of the conclusions.

9 Conclusion

This study benchmarks four representative segmentation systems and seven ablations on Pascal VOC 2007 under a common, validation-driven training pipeline. SegFormer-B2 is the strongest model by the primary overlap metrics, DeepLabV3+ is the strongest CNN baseline and the best HD95 model, and SAM2 becomes competitive only after lightweight LoRA adaptation. The ablation suite shows that encoder capacity, augmentation strength, and training budget all matter materially, while the default cross-entropy-plus-Dice loss remains a sound compromise. More broadly, the project demonstrates that even a course-scale benchmark can support a meaningful empirical paper if the evaluation protocol is locked, the figures are interpreted carefully, and the conclusions are stated at the level the evidence actually supports.

A Additional Qualitative Panels

Figure 7 shows the best and worst SegFormer-B2 triptychs on the official validation split. The best cases are dominated by large, visually distinctive objects with clean boundaries or high foreground occupancy. The worst cases frequently contain thin structures, severe occlusion, or confusing boundaries between adjacent classes. This asymmetry is a useful reminder that the mean metrics are still influenced heavily by a relatively small number of hard examples.

Figure 8 focuses on person-present failure cases. Person masks are difficult because they combine thin limbs, occlusion, and context-dependent confusion with bicycles, motorbikes, and indoor background clutter. Even the best model frequently localizes the person coarsely before losing fine structure at arms, legs, and object contact points. That recurring pattern mirrors the lower person-subset scores in Figure 4.

Figure 8: SegFormer-B2 failure cases restricted to person-present images. The dominant errors arise at articulated limbs, occlusion boundaries, and interactions with nearby objects.

References

Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *European Conference on Computer Vision*, pages 833–851, 2018.

Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. The pascal visual object classes (voc) challenge. *International Journal of Computer Vision*, 88(2): 303–338, 2010.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Borja Mañana, Swetadri Saha, Quanfu Hu, Dikai Wu, Manohar Paluri, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.

Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention*, pages 234–241, 2015.

Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. In *Advances in Neural Information Processing Systems*, volume 34, pages 12077–12090, 2021.